

Report R14 — one figure for the WCC/SCC planes and the flow between them

Candidate layouts

Tier 1 analysis

2026-08-04

R9 spends three figures on a single idea: every rule has a position in the sector plane, a position in the monitored-attractor plane, and dissipation moves it from the first to the second. Two candidates follow, then the three first-pass layouts they replaced.

The fact that shapes both. Of the 230 rules that move, 220 move *straight down* and all 230 decrease b ; not one moves horizontally. That is forced rather than empirical: one attractor per sector means $n_{\text{sc}} = n_{\text{wc}}$, so $a=a$ and the move can only be vertical. A horizontal component therefore exists *if and only if* the rule has a sector holding more than one attractor — R9 §7.3's twelve, of which ten show it (rules 235 and 249 have a constant ratio of 2, which cancels out of a growth base). So the ten off-axis arrows are not a scatter of outliers; they are exactly the multi-attractor rules, and both candidates below draw only those.

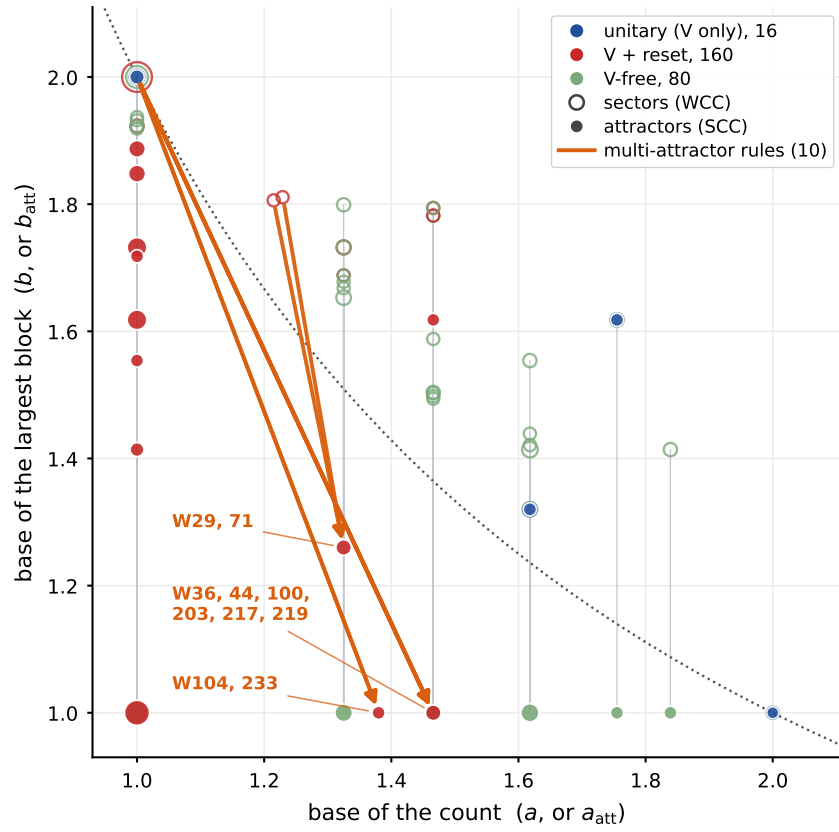


Figure 1: **A** — **overlay**. One plane. Each rule appears twice, hollow where its sectors put it and filled where its attractions do. The 220 forced-vertical moves are hairline stems with no arrowhead — they carry no information the paired markers do not already give — and only the ten *exceptions* are drawn as arrows, labelled. Most compact of the layouts, and the only one in which both coordinates live on the same axes.

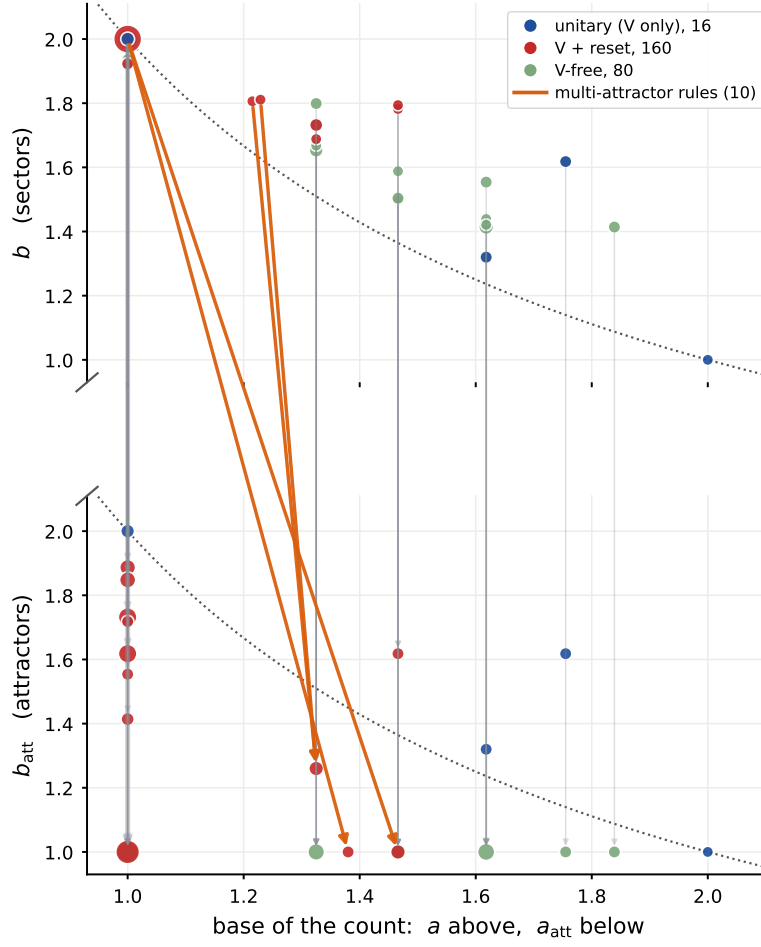


Figure 2: **B** — **stacked, one shared x -axis, split y** . The x -coordinate means the same thing in both planes, so it is drawn once, at the bottom. The two y -axes are separate, marked by the break slashes on the left spine, and the gap between the panels is left completely empty — no spine, no ticks, no panel titles — so an arrow never crosses a line or a label. Because $a=a$ for every rule but ten, almost every arrow is then exactly vertical and the ten that lean are the multi-attractor rules (orange).

First pass, for the record

These three were tried first and are kept only to show what was rejected: C and D put the two planes in separate coordinate systems, so the eye cannot compare positions and the verticality of the flow is invisible; E buries it entirely.

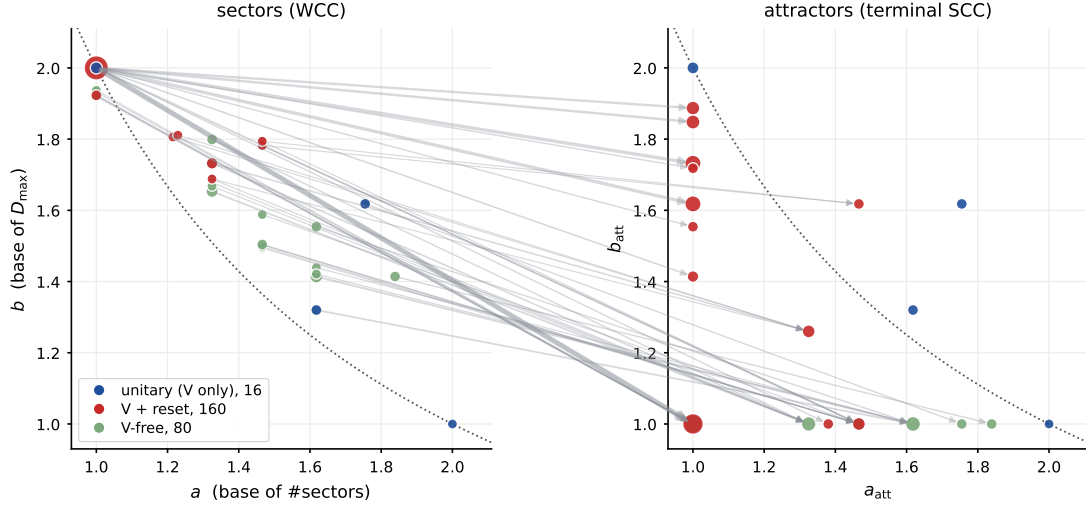


Figure 3: **C** — twin panels, horizontal. Arrows bundle in the gutter and the gap costs a third of the width.

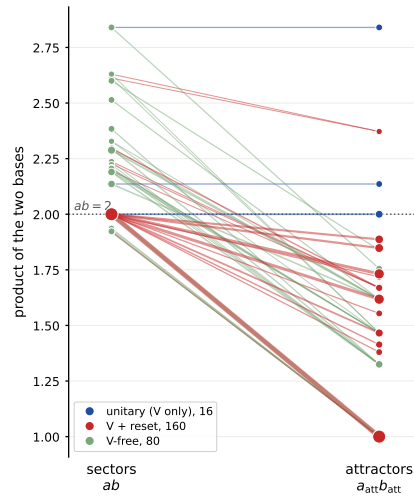


Figure 4: **D** — **slopegraph** of the product ab alone. Discards the a/b split but makes the exclusion curve the picture: everything starts on or above 2, the unitary rules run exactly horizontal, and almost everything else drops below. Still useful as a small companion panel.

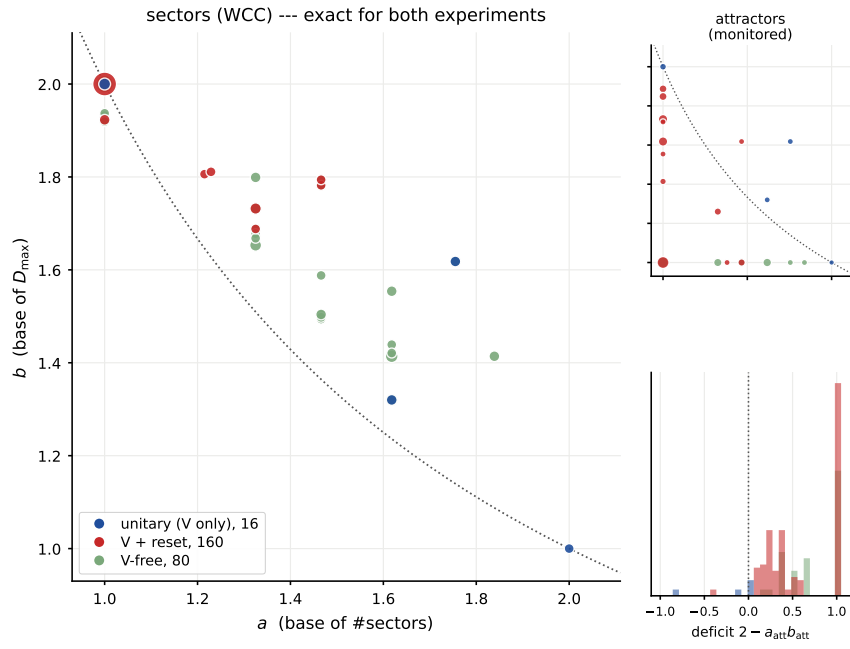


Figure 5: **E** — **inset**. Reflects the asymmetry of the two maps — the sector plane is exact for both experiments, the attractor plane is monitored-only — but loses the flow altogether.

Recommendation

A for a single-panel figure, **B** when the two planes must stay visually distinct, and **D** as a narrow companion to either if the $ab \geq 2$ theorem needs to be shown as well.

A and B encode the same claim in two ways: the flow is one-dimensional except for ten rules, and those ten are exactly the ones whose sectors carry several attractors. A says it by superposing the planes; B says it by stacking them on a common x so that “vertical” is literal.

Reproduce

```
python -m qca_fragmentation.viz.journal --bc obc0
```

writes all layouts as `figures/fig-journal_{style}-obc0.{pdf,png}`; the module is `code/qca_fragmentation` and `journal.off_axis()` returns the ten exception rules.